Google Cloud H2S

AI Builder Cup

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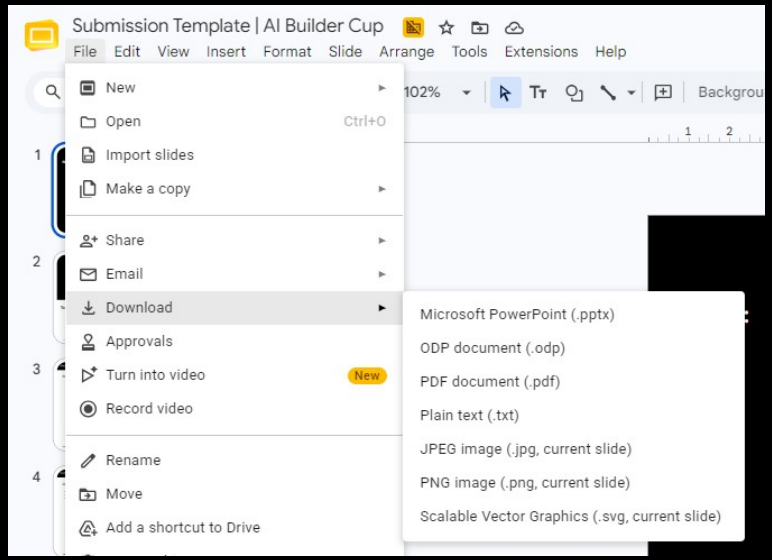
OpsMind — AI Operational Intelligence

Google Cloud AI Builder Cup 2026 · JAPAC

Team: ABB Catalyst · Team lead: Alwi Hamzah

Problem: operations data is split; a power spike does not obviously track production.

Prototype: deterministic metrics + Gemini explanations. AI suggests; a person decides.



Executive summary

Operations data is split across production, electricity, hours and weather.

OpsMind calculates intensity, a trailing baseline, deviation and a potential cost estimate.

Gemini explains only the supplied evidence and lists what is still missing.

The model never calculates the numbers. Every demo row is labelled SIMULATED DATA.

AI suggests; a person decides.

Brief about the idea

- OpsMind is an AI Operational Intelligence prototype for SME manufacturers.
- It computes energy intensity (kWh/unit), a trailing baseline, deviation and a potential excess-cost estimate.
- Gemini explains the evidence, separates facts from hypotheses and recommends human checks.
- All demo data is labelled SIMULATED DATA. No saving is claimed.

Opportunities

- Differentiation: evidence-first investigation, not a generic chatbot or a single threshold alert.
- Every observation cites evidence E1–E13; each hypothesis lists the evidence that is still missing.
- The cost panel shows the formula and assumptions — not a claimed bill or verified saving.
- Weather is context, not proof; before/after changes are reported without asserting causation.
- **USP: a human-in-the-loop path from detection → investigation → action → verification.**

Features

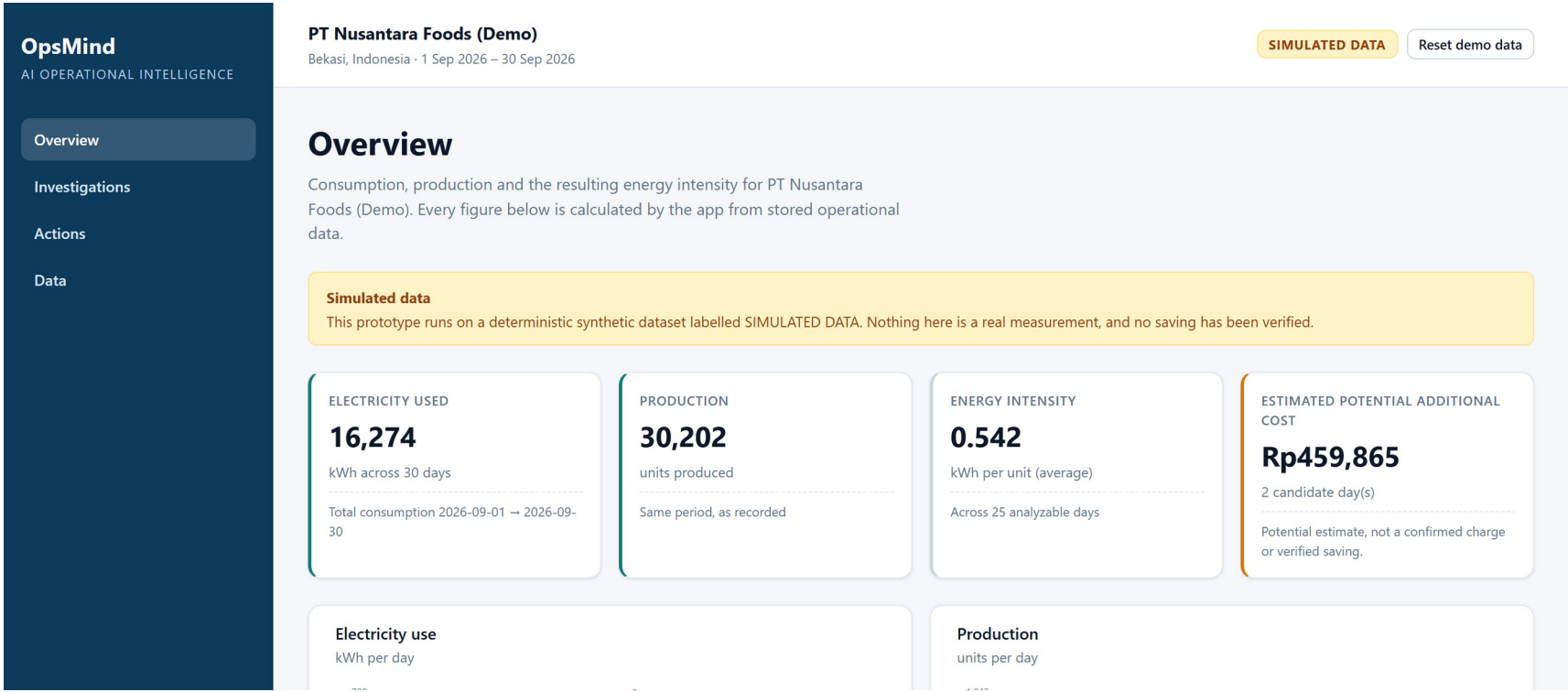
- Daily dashboard: production, electricity, energy intensity and potential additional cost.
- Transparent detection: trailing baseline, 15% heuristic threshold and severity bands.
- AI investigation: Gemini or a labelled deterministic mock; Zod-validated output.
- Evidence: 13 items (E1–E13), facts vs hypotheses, missing inputs and limitations.
- Action workflow: Open / In progress / Completed with timestamps and completion notes.
- Before/after comparison with an explicit “observed change only” caveat.

Investigation flow



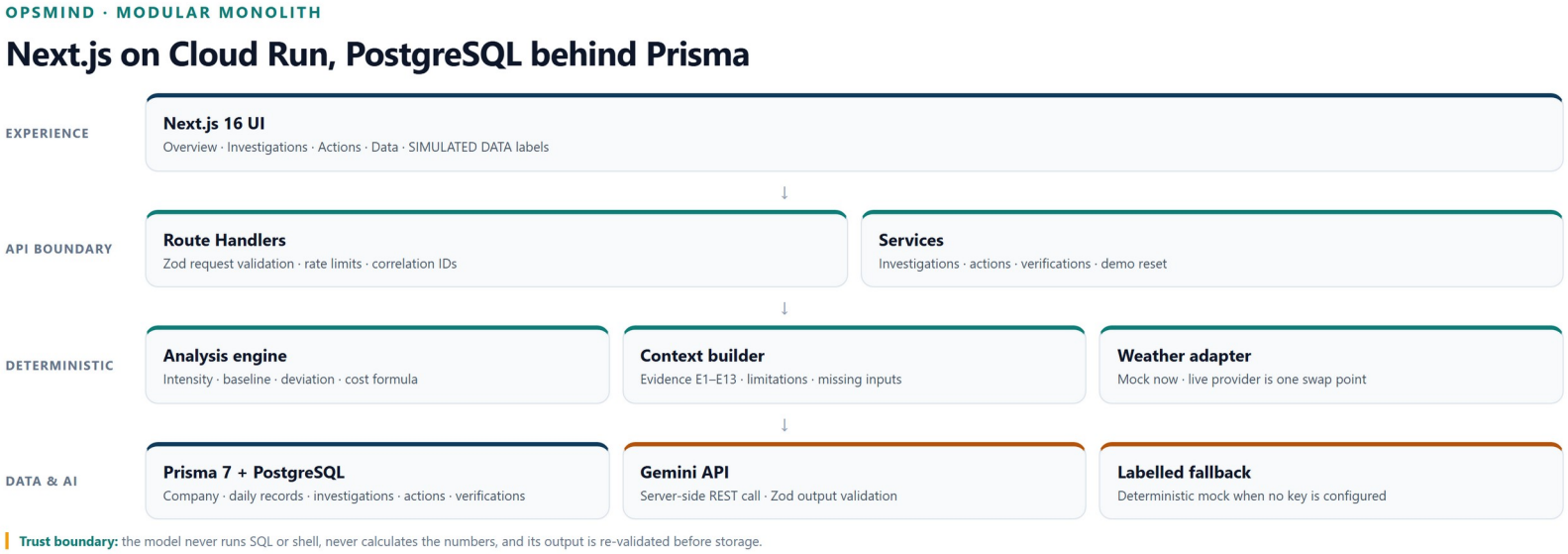
The deterministic engine calculates; Gemini explains; a person decides. Model failure is stored as FAILED.

Product walkthrough — Overview



30 simulated days · 17 Sep 2026: +36.0% vs the trailing baseline · potential estimate Rp288,884.

Architecture



Next.js → route handlers → validation → services → PostgreSQL; Gemini output is validated before storage.

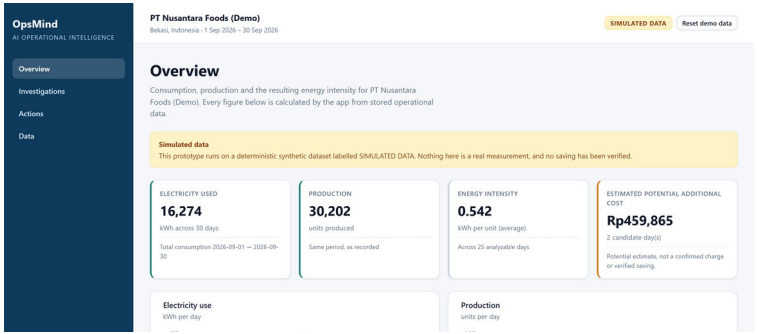
Technology stack

- Next.js 16 App Router · React 19 · TypeScript 7 strict · custom CSS design tokens.
- PostgreSQL + Prisma 7 driver adapter · native decimal and date columns.
- Gemini API via server-side REST · Zod 4 output validation · labelled deterministic mock fallback.
- Vitest 5 (79 tests) · Playwright E2E · Docker · GitHub Actions.
- Local embedded PostgreSQL · deployment target: Google Cloud Run.

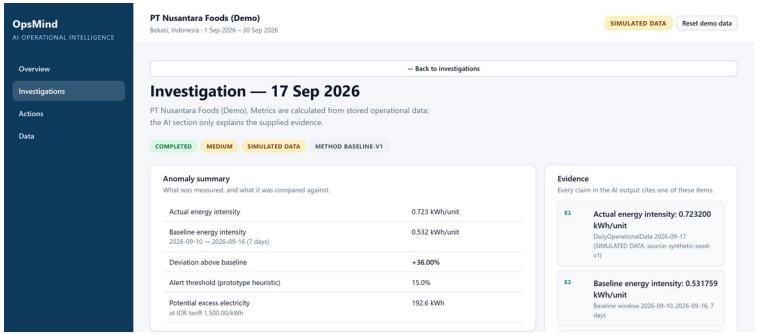
Estimated implementation cost (optional)

- Development: local embedded PostgreSQL + labelled mock investigator; no cloud spend required.
- Deployment: Cloud Run scale-to-zero + managed PostgreSQL + Gemini token usage; cost depends on traffic and region.
- The app reports a potential estimate: excess kWh × tariff, with the tariff and assumptions shown.
- It does not claim verified savings; a controlled before/after measurement is required.

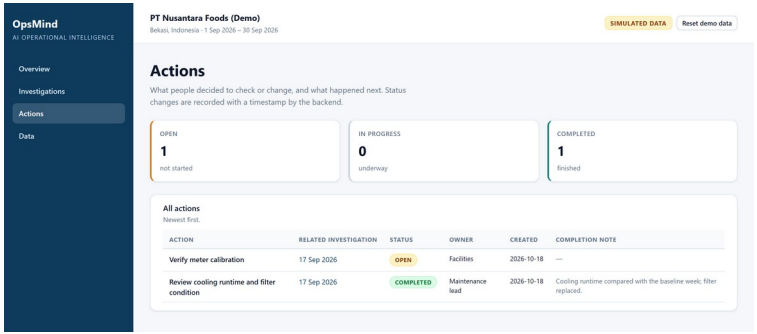
Prototype snapshots — simulated data



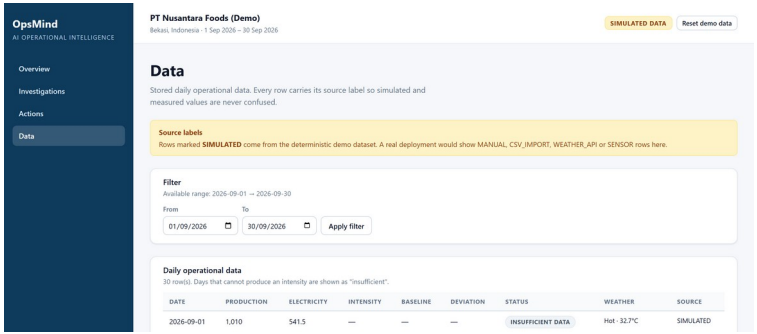
Overview — KPIs, trend and investigation queue



Investigation — evidence E1–E13 and AI explanation



Actions — open/completed and before/after comparison



Data — source labels and date filter

Performance and verification

- 79 automated tests: calculation engine, validation, AI failure modes and API integration.
- 3 Playwright E2E tests: overview, sidebar navigation and the investigate journey.
- Typecheck and production build pass; the 30-day demo dataset is deterministic.
- Health check: /api/health reports database and AI-provider status.
- Not claimed: production throughput, a latency benchmark, or real savings.

Roadmap

- Real weather provider with coverage, terms, attribution and rate-limit review.
- CSV upload with preview/validation; multi-site support.
- Authentication/RBAC and gateway-level rate limits.
- Controlled before/after studies with measured savings.
- Model evaluation on real operations data; prompt/model versioning.

Submission links

- GitHub public repository: see the Hack2skill submission form.
- Demo video (<3 min): see the submission form.
- Live prototype: see the submission form.
- Health check: /api/health (database + AI-provider status).
- **The submission form is the source of truth for the final URLs.**

OpsMind — AI Operational Intelligence

Ask: what is the evidence?

Demo candidate: 17 Sep 2026, +36.0% above the trailing baseline — a potential estimate, not a verified saving.

AI suggests; a person decides.

Thank you.

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Thank you

